

DRDO MALE UAV PROPULSION FMECA REPORT (PS-26054)

Failure Mode, Effects, and Criticality Analysis — Rotax 912 iS Sport Aero-Engine

1. FMECA SUMMARY & RISK PRIORITY NUMBERS (RPN)

Fault ID	Component	Failure Mode	Severity (S)	Occurrence (O)	Detection (D)	RPN	Primary Causal Pathway
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01	Shroud Baffle	Seal Degradation / Leak	8	4	2	**64**	Baffle leak -> Cyl 2 cooling air deficit -> CHT2 surge -> Oil heating -> Radiant heat to Cyl 4
02	Injector #1	Nozzle Clog / Cavitation	9	3	2	**54**	Nozzle restriction -> Lean burn Cyl 1 -> EGT1 spike -> Torque ripple -> RPM sag & vibration
03	Lane A Coil	Spark Lead Breakdown	7	4	2	**56**	Spark loss -> Unburnt fuel -> EGT2 drop -> Cyclic misfire -> RPM flutter & gearbox shock
04	Oil System	Scavenge Cavitation / Relief	9	3	2	**54**	Pressure drop -> Bearing friction surge -> Oil temp spike -> Block heat to all heads -> RPM drag
05	Gearbox	Dog-Clutch Micro-Pitting	7	4	2	**56**	Tooth pitting -> 3rd harmonic resonance -> Parasitic drag -> RPM sag -> MAP rise & oil heating
06	Exhaust #3	Mixture Disparity / Gasket	6	4	2	**48**	Mixture lean out -> EGT3 spike -> CHT3 warming -> Asymmetric exhaust pulse & mild vibration
07	Alternator	Stator Thermal Sag / Slip	6	4	2	**48**	Generation deficit -> Bus voltage sag -> Battery discharge -> Weak coil dwell & RPM sag
08	FADEC ECU	MAP Transducer Drift	7	3	2	**42**	MAP drift high -> Speed-density overfueling -> Fuel flow rise -> Rich quench all EGTs cool

2. CROSS-SUBSYSTEM COUPLING LAWS

- Thermal-Mechanical Coupling:** Hydrodynamic bearing starvation directly converts mechanical torque loss into oil thermal energy ($\Delta T_{oil} \propto RPM \times \Delta P_{oil}^{-1}$).
- Combustion-Kinematic Coupling:** Uneven single-cylinder torque stroke production creates crankshaft rotational velocity fluctuation ($\Delta RPM_{flutter} \propto \Delta EGT_i$).
- FADEC Closed-Loop Dynamics:** In speed-density injection, any unmeasured pressure drift directly biases fuel air-charge delivery ($m_f \propto MAP_{reported}$), altering all downstream exhaust temperatures.